University of Global Village (UGV), Barisal

Experiment list of Sessional Course

Winter-2026

Name of Subject: Electronics-I sessional / Principle of Electronics **Department: EEE**

Subject Code: EEE-0714-1202 **Semesters: CSE 3rd, EEE 2nd**

Section: CSE: A & B. EEE: A & A

Group: CSE-A, EEE-A&B **Session:**

Credit: 01

Student Name: **Student ID:**

SL No:	Title of Practical Skills	Lab Topic	Learning Outcome	Instructor Signature only for Competent
1	Study of P-N Junction Diode VI Characteristics	Measurement of voltage and current in forward and reverse bias configurations for a silicon diode.	Students will understand the concept of knee voltage and potential barriers. They will also learn to calculate static and dynamic resistance from the V-I curve.	
2	Characteristics of Zener Diode and Its Application as a Voltage Regulator	Analysis of Zener breakdown voltage and the diode's ability to maintain a constant output under varying loads.	Students will identify the difference between Zener and conventional diodes. They will also learn to design a shunt voltage regulator using Zener breakdown properties.	
3	Design and Performance Analysis of Half-Wave and Full-Wave Rectifier Circuits	Construction of single-diode and center-tapped transformer circuits to convert AC to pulsating DC.	Students will compare the efficiency and ripple factors of different rectification methods. They will also learn to visualize unidirectional current flow on an oscilloscope.	

4	Design and Performance Analysis of a Bridge Rectifier Circuit	Implementation of a four-diode bridge configuration to achieve full-wave rectification without a center-tapped transformer.	Students will analyze the Peak Inverse Voltage (PIV) requirements for bridge diodes. They will also learn how filter capacitors reduce output ripple.	
5	Study of Clipper and Clamper Circuits	Circuit design for wave-shaping by removing specific portions of a signal or shifting its DC level.	Students will understand how to limit signal amplitudes using diode clipping. They will also learn to add DC offsets to AC signals using clamping circuits.	
6	Study of BJT as a Switch	Investigation of Bipolar Junction Transistors operating in saturation and cut-off regions to control a load.	Students will determine the base current required to drive a transistor into saturation. They will also learn to use BJTs for digital switching applications.	
7	Study of MOSFET as a Switch	Analysis of N-channel MOSFETs used as high-speed electronic switches for power control.	Students will understand the relationship between gate-source voltage and drain-source resistance. They will also learn the advantages of MOSFETs in low-power dissipation switching.	